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Digital Twin Applications in EPC Projects

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Abstract

Engineering drawings are the most critical project documentation, which provides processing plant blueprint. Relational database and attributes driven graphics are being used to transform CAD drawing to intelligent drawings. These intelligent drawings and design data is used to make smart decisions. The latest methods to create project digital twin are evolving to make use of these intelligent drawings. This paper discusses the user friendly and innovative solutions to accomplish digital twin goals at EPC stage.

P&ID is a valuable data source for oil and gas projects. It is used to generate pipelines, instrumentations and equipment lists. These are mainly used for material take-off and other procurement purposes. In Intelligent P&IDs these lists reporting may be customized and attained, automatically. During digital twin creation process, sharing of data with different departments via direct links and data lake is necessitated. During project EPC stage, data lake may be used to find many out of the box solutions e.g., training AI models to predict project completion dates and material supply-demand issues, proactively.

Due to stringent project schedules, it is always challenging to achieve an important milestone of timely issuance of Intelligent P&IDs. Variety of difficulties during development of intelligent P&IDs are summarized below:

1. Drafting of modification and demolition P&IDs for brown-field project
2. Drafting of colour coded system turnover P&IDs
3. Revision & scope clouds Customization to meet digital twin requirements methods in available intelligent software are not user friendly.

Many steps are involved to perform above mentioned tasks. Various lists generated from P&IDs such as line list consumes significant time and prone to error if not managed properly. Digital twin requirements make it further complex. Various techniques are used to perform routine drafting of P&IDs, attain required reporting and digital twin targets. However, these need to be managed and customized by highly skilled professionals. There is a need for tailor-made solutions in the intelligent P&ID software to make operations user friendly. Cloud computing big data handling capability and AI may help to meet digital twin requirements. Archiving and retrieving data from ultra-fast cloud facilities may solve many problems.

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This paper provides ideas to debottleneck various limitations in the currently available drafting software. It is to add more intelligence and smooth interconnectivity among different disciplines thus making them more user friendly and smart. This all may translate in speeding up the drafting and error free reporting with the help of AI. It may enhance software capabilities beyond simple drafting and reporting to meet new era requirements of digital twin.

Evolution of CAD (Computer Aided Design) and Digital Twin

There are many types and kinds of engineering drawings e.g., plant schematics, logical diagrams (P&IDs, PFDs, MSDs and electrical / instrumentation diagrams), orthographic projections drawings (scaled and dimensioned layouts, plot plans, alignment sheets, fabrication cut lists, hoppers and ducts development drawings, mechanical assemblies and related detailed drawings), isometric and cavalier projection drawings.

CAD (Computer Aided Design) software has been in the market for a long time, but after 1990 it started replacing conventional drafting boards and machines, very speedily. The emergence of PCs (Personal Computers) and GUI (Graphical User Interface) atmosphere has also played a major role in giving rise to CAD software. Based on GUI philosophy, Windows and Macintosh operating systems have a major role in making PCs user friendly and easy to operate. Word processing and calculation spreadsheet software indirectly helped CAD users to get better and more accurate results, robustly. CROs (Central Record Offices) within any EPC or engineering design company required a large physical space to store drawings in an old Victorian era library setup. Modern computer hard disk drives and dedicated servers helped to store a very large quantity of drawings and documents in a very small space. This all helped us to design and manage relatively larger projects than ever before.

Twenty first century started with the great influence of computers in our daily lives as well as in the energy sector. PCs started playing major roles in all spheres of professional lives due to the enhancements in computer hardware systems, improved GUI OSs (Operating Systems), rise of easy computer coding languages and the advent of internet. This all also has opened the doors for smart CAD (2D and 3D both). RDB (Relational Database) and attributes driven 2D and 3D graphics are being used to create smart drawings. Smart CAD helped to build and manage further big processing plants in the energy sector.

Processing plant sizes are getting bigger day by day and it is equally multiplying and ballooning design and engineering activities. Local servers and hard disks are getting replaced by hi-class cloud services to handle the emerging demand of robust computing and bigger disk space for computing, archiving and retrieval. In today's corporate world, the CPUs (Central Processing Units) and TBs (terabytes) disk space is insufficient to do the said tasks nicely. Scalable, reliable, secure, efficient, cost effective and fault tolerant cloud services are easily available now. GPUs (Graphics Processing Units) or TPUs (Tensor Processing Units) and PTs (petabytes) large disk space is available on demand. More computing power and theoretically infinite storage space is available now to handle big data (filled with drawings, documents and structured or unstructured information). In pure financial and business terms, it has significantly cut COPEX (Capital Expenditures) and OPEX (Operating Expenditures) costs. Time span has reduced to minutes from days to deploy these tools.

Now, methods to create project digital twin are evolving. In a multidisciplinary design and engineering atmosphere, RDB, data driven graphics and cloud computing and storage powers are helping to integrate and analyze inputs, outputs, and results to achieve digital twin goals. Fusion of AI and Smart CADDING is the new front to achieve further big and great.

All engineering drawings and documents are important, but P&ID has a primary role during this whole activity, which provides foundations to design, construct and operate a processing plant. One of the main purposes of digital twin is to build a real-world processing plant's digital replica. P&ID remains the main document during this exercise. It is P&ID which gives directions to piping department to develop accurate 3D models and prepare MTO (Material Takeoff reports). The Instrumentation department is bound to get

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directions from P&ID to develop its detailed drawings. Mechanical department fetch important design data of equipment from P&IDs to move forward. Commissioning related testing is not possible without systems turnover P&IDs in hand. In short, from concept to operations, running plant surveys, cost calculations and plant brown filed revamping activities we need P&ID as a basic document.

In this paper we are critically analyzing available P&ID applications limitations and their role in digital twin model. This paper is primarily and critically analyzing smart P&IDs role in digital twin applications, but collective achievements and fears supported with concrete recommendations are also shared for the subject cause i.e., Digital Twin Applications in EPC Projects.

How digital twin applications may play their role during any processing plants average life of 30 to 50 years? It is a big question which demands answers with concrete recommendations and solutions. This paper has critically analyzed this issue also and has proposed some recommendations to forestall this futuristic problem in a more plausible and preemptive style.

Smart P&ID Role in Digital Twin

During the pre-digital twin era, smart drawings and 3D models have produced remarkable results. Still, we are getting good results, but most of the time those results were compiled and stacked in individual siloes. Unique tagging conventions for piping components, instruments, pipelines, and equipment are used throughout the project (from design till operation). These are not specific to individual disciplines e.g., process, piping, instrument or mechanical but application to whole project. Somewhat quasi-centralized interdisciplinary integration worked well during smart CADDING era, but now it is strongly realized to centralize the flow of unique tags to streamline multidisciplinary EPC activities. It may help to save manhours by avoiding repeated tasking, duplicity detection and may improve overall project management and performance.

After enumerating some accomplishments, this paper is highlighting software limitations and attributes driven graphics preservations issues in the future.

Achievements

Digital twin platform is launched successfully. This is helping EPC contractors as well as esteemed clients. There are several applications involved to accomplish the goal, but the paper's primary focus is on Smart P&IDs. Data is flowing from P&ID to multiple directions to benefit all involved. Through XML export, our digital twin team has integrated different vendors utilities output to get roust and better results for digital twin model.

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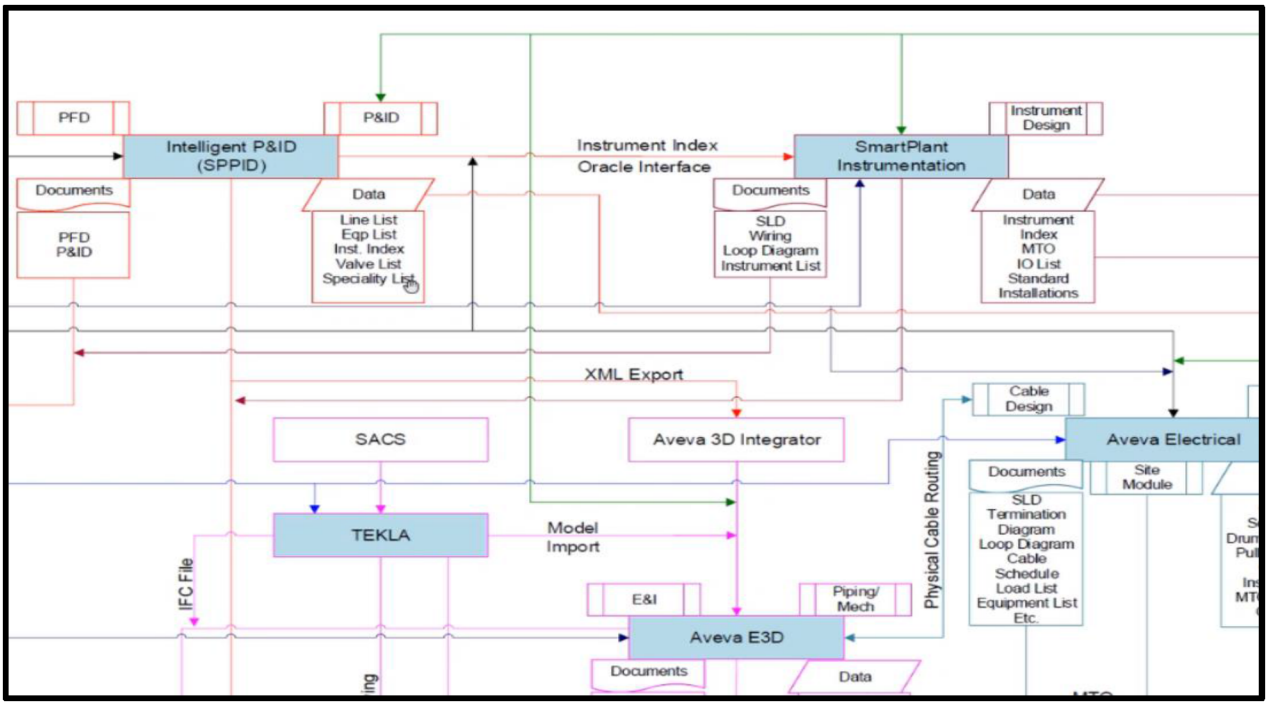

Under digital twin umbrella some smart P&IDs products are listed below:

1. Conventionally MTO (material takeoff) report is generated by piping department in EPC setup. NPCC digital twin team has successfully automated an innovative MTO report from P&ID. Related standards are interlinked with certain piping components at the backend and we are able to extract MTO from P&ID. This report excludes pipeline length and elbows count only. MTO on P&IDs stage is very helpful to have an overall idea and it may also be used to cross check MTO extracted from 3D model during next stages and phases of 3D modeling.
2. During the design and engineering phase of brown field project there is a need to identify new and existing areas. Vendor scope should also be clearly marked. Graphically we can mark and identify these areas in any drawing, but once we are going to get data extracted intelligently then it's not that easy of a task. In collaboration with the process engineering department, our digital twin team made it possible to intelligently extract a line list report, which is also taking care of new, existing, future and vendor scopes.
3. Future facilities are normally shown in P&IDS with dotted lines but during intelligent valve report extraction there is need to identify future fitting and valves etc. In collaboration with the process engineering department, NPCC digital twin team has successfully configured a intelligently extractable valve list which is also taking care contractor / vendor scopes.

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Primary Dwg No	Secondary Dwg No	LineNo	Secondary Drwg Rev No.	Abbreviation	Designer Size1	Client Class Code	MTO Qty Hold?	ClassDesc	Takeoff quantity	Description
ZTPS-4-4"-BD-01109-15LE1P11	CA-223722001	4"-BD-01109-15LE1P11	A	VGST	1	15SV0P	0	CL1500	1	Gate valve
ZTPS-4-4"-BD-01109-15LE1P11	CA-223722001	4"-BD-01109-15LE1P11	A	PL	1	15SV0P	0	CL1500	1	Plug
ZTPS-4-1"-BD-01114-1SV0P	CA-223722001	1"-BD-01114-1SV0P	A	F15	1	1SV0P	0	CL1500	1	Flange
ZTPS-4-1"-OS-03001-1SV0P	CA-223722001	1"-OS-03001-1SV0P	A	F15	1	1SV0P	0	CL1500	1	Flange
ZTPS-4-1"-OWS-04701-1SV0P	CA-223722001	1"-OWS-04701-1SV0P	A	F	1	1SV0P	0	CL150	1	Flange
ZTPS-4-1"-OWS-04701-1SV0P	CA-223722001	1"-OWS-04701-1SV0P	A	FB	1	1SV0P	0	CL150	1	Blind flange
ZTPS-4-12"-P-01100-15LE1P13	CA-223722001	12"-P-01100-15LE1P13	A	PL	1	15SV0P	0	CL1500	1	Plug
ZTPS-4-12"-P-01100-15LE1P13	CA-223722001	12"-P-01100-15LE1P13	A	VGST	1	15SV0P	0	CL1500	1	Gate valve
ZTPS-4-12"-P-01100-15LE1P13	CA-223722001	12"-P-01100-15LE1P13	A	VG	1	15SV0P	0	CL1500	1	Gate valve
ZTPS-4-12"-P-01100-15LE1P13	CA-223722001	12"-P-01100-15LE1P13	A	VGST	1	15SV0P	0	CL1500	1	Gate valve
ZTPS-4-12"-P-01100-15LE1P13	CA-223722001	12"-P-01100-15LE1P13	A	PL	1	15SV0P	0	CL1500	1	Plug
ZTPS-4-12"-P-01100-15LE1P13	CA-223722001	12"-P-01100-15LE1P13	A	VGST	1	15SV0P	0	CL1500	1	Gate valve
ZTPS-4-12"-P-01100-15LE1P13	CA-223722001	12"-P-01100-15LE1P13	A	PL	1	15SV0P	0	CL1500	1	Plug
ZTPS-4-12"-P-01100-15LE1P13	CA-223722001	12"-P-01100-15LE1P13	A	VG	1	15SV0P	0	CL1500	1	Gate valve
ZTPS-4-12"-P-01101-15LE1P11	CA-223722001	12"-P-01101-15LE1P11	A	VG	1	15SV0P	0	CL1500	1	Gate valve
ZTPS-4-12"-P-01101-15LE1P11	CA-223722001	12"-P-01101-15LE1P11	A	VG	1	15SV0P	0	CL1500	1	Gate valve
ZTPS-4-12"-P-01101-15LE1P11	CA-223722001	12"-P-01101-15LE1P11	A	VG	1	15SV0P	0	CL1500	1	Gate valve
ZTPS-4-12"-P-01101-15LE1P11	CA-223722001	12"-P-01101-15LE1P11	A	VG	1	15SV0P	0	CL1500	1	Gate valve
ZTPS-4-4"-P-01102-15LE1P11	CA-223722001	4"-P-01102-15LE1P11	A	PL	1	15SV0P	0	CL1500	1	Plug
ZTPS-4-4"-P-01102-15LE1P11	CA-223722001	4"-P-01102-15LE1P11	A	VGST	1	15SV0P	0	CL1500	1	Gate valve
ZTPS-4-4"-P-01102-15LE1P11	CA-223722001	4"-P-01102-15LE1P11	A	VGST	1	15SV0P	0	CL1500	1	Gate valve

Plant Group Name	Line Number	Size (Nominal Bore)	Piping Spec/ Class	Fluid Code	From	To	Sheets	Status
CM	06011	1"	1SD0P	AS	A90-CM-D-0004	TO BOAT LANDING	CA-229454014	New
CM	08021	1"	1SD0P	AS	HEADER	A90-CM-D-0004	CA-229454014	New
CM	01002	4"	15LE1P12,90LE1T	P	WELLHEAD #1322	16"-P-02001-15LE1P12	CA-229454012,CA-229454002	New
CM	01003	4"	15LE1P12,90LE1T	P	WELLHEAD #1323	16"-P-02001-15LE1P12	CA-229454003,CA-229454012	New
CM	01004	4"	15LE1P12,90LE1T	P	WELLHEAD #1324	16"-P-02001-15LE1P12	CA-229454004,CA-229454012	New
CM	01005	4"	15LE1P12,90LE1T	P	WELLHEAD #1325	16"-P-02001-15LE1P12	CA-229454012,CA-229454005	New
CM	01006	4"	15LE1P12,90LE1T	P		16"-P-02001-15LE1P12	CA-229454006,CA-229454012	Future
CM	01007	4"	15LE1P12,90LE1T	P	WELLHEAD #1327	16"-P-02001-15LE1P12	CA-229454007,CA-229454012	New
CM	01008	4"	15LE1P12,90LE1T	P	WELLHEAD #1328	16"-P-02001-15LE1P12	CA-229454008,CA-229454012	New
CM	01009	4"	15LE1P12,90LE1T	P	WELLHEAD #1320	16"-P-02001-15LE1P12	CA-229454012,CA-229454009	Future
CM	01010	4"	15LE1P12,90LE1T	P	WELLHEAD #1329	16"-P-02001-15LE1P12	CA-229454012,CA-229454010	New
CM	01011	4"	15LE1P12	P	4"-P-01001-90LE1T	6"-P-02010-15LE1P12	CA-229454001,CA-229454012	New
CM	01012	4"	15LE1P12	P	4"-P-01002-90LE1T	6"-P-02010-15LE1P12	CA-229454012,CA-229454002	New
CM	01013	4"	15LE1P12	P	4"-P-01003-90LE1T	6"-P-02010-15LE1P12	CA-229454003,CA-229454012	New
CM	01014	4"	15LE1P12	P	4"-P-01004-90LE1T	6"-P-02010-15LE1P12	CA-229454004,CA-229454012	New
CM	01015	4"	15LE1P12	P	4"-P-01005-90LE1T	6"-P-02010-15LE1P12	CA-229454005,CA-229454012	New

Line ID	P&ID Name	Nominal Diameter	Type	Piping Materials Class	Opening Action	Construction Status	Supply By	Remarks
20"-P-01509-3CS1P	RA-040465001	20"	Ball valve	3CS1P		New	By Vendor	
1"-N-01474-1CS1P	RA-040474001	1"	Globe valve	1CS1P		New	By Contractor	
20"-P-01570-3CS1P	RA-040464001	1/2"	Gate valve	3CS1P	NC	New	By Vendor	
2"-AS-01757-1NT1A	RA-040475001	1"	Gate valve	1NT1A	NC	New	By Contractor	
1/2"-N-01475-1CS1P	RA-040473001	1/2"	Gate valve	1CS1P		New	By Contractor	
20"-P-01511-3CS1P	RA-040466001	3/4"	Gate valve	3CS1P	NC	New	By Contractor	
20"-P-01512-3CS1P	RA-040464001	20"	Check valve	3CS1P		New	By Vendor	
8"-CDH-01633-1CS1P	RA-040474001	8"	Ball valve	1CS1P	LO	New	By Contractor	FB
54"-P-01507-3CS1P	RA-040465001	1/2"	Gate valve	3CS1P	NC	New	By Vendor	
20"-P-01511-3CS1P	RA-040466001	3/4"	Gate valve	3CS1P	NC	New	By Contractor	
3/4"-OWS-01496-1BD0U	RA-040472001	3/4"	Gate valve	1BD0U		New	By Contractor	
20"-P-01510-3CS1P	RA-040466001	3/4"	Gate valve	3CS1P	NC	New	By Contractor	
20"-P-01570-3CS1P	RA-040466001	3/4"	Gate valve	3CS1P	NC	New	By Contractor	
20"-P-01511-3CS1P	RA-040466001	3/4"	Gate valve	3CS1P	NC	New	By Contractor	
3"-CDH-01638-1CS1P	RA-040475001	3"	Ball valve	1CS1P	LO	New	By Contractor	

Limitations of Smart P&IDs

P&ID is a major document embedded in the deep roots of any digital twin model of any processing plant. Maximum data flow within digital twin and related applications form P&ID is like pumping healthy blood in the body of any digital twin model. During materialization of Digital Twin goals and in the past certain

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limitations have been observed in available smart P&ID software. We will highlight a few in the paper. P&ID is a main and primary document in any processing plant:

1. Brown field P&IDs (construction and demolition P&IDs) handling is not easy and user-friendly task in available smart P&ID software. Graphics and data extraction both have lacunas and discrepancies, which need to be mitigated. Demolition and construction P&IDs names are normally the same.

    a. How to keep 02 files with same name in same database together? Available smart P&ID software are unable to address this issue wisely.
    b. Demolition P&ID is dumb and construction P&ID is smart. Digital Twin demands maximum stuff available on same platform not in siloes. Demolition VS Construction P&IDs case directly challenges this objective. Available smart P&ID software are unable to address this issue wisely.
2. During brown field project engineering, EPC companies are bound to their scope area only. EPC companies control / identify their scope area with scope clouds mostly. Smart P&ID software is aesthetically and intelligently very poor to address this issue.
3. Line list is a main report extracted from smart P&ID software. We get results, but the process is very cumbersome, awkward and not user-friendly. Intelligent line list should be able to include / exclude future, new, revamped and vendors scopes by itself and automatically.
4. Instrument list is a very critical document, which inherit data from P&ID for multiple disciplines use and even for operations It is utilized by instrumentation, construction, commissioning and operations departments on different stages to accomplish their tasks, respectively. For example, during drafting a short detailed graphics of MOV, PCV & FCV etc. are shown in the P&IDs. Comprehensive detail is implied and hidden somewhere in invisible space, but it is available in legends. The instrumentation list should carry whole details of each implicitly or explicitly available instrument. Only an admin with high level smart P&ID admin skill may configure required reporting system, which is not good.

Software Enhancement Proposals

Any smart P&ID admin with years of working experience, IT skills & dexterity, handful process engineering knowledge and understanding of P&ID may only configure an excellent reporting system and mechanism. Proposals are listed below to handle above stated issues:

1. To render a good and realistic digital twin model, demolished and constructed areas details (both 2D and 3D) during plant life cycle should be available at the disposal of simple buttons clicks. It will help operating companies to conduct several studies successfully in the proceeding years of plant operation and its life.
2. Revision and scope clouds are not up to the mark in available smart P&IDs software. May times they overlap or form an awkward shape, once drawn. Drawing them in smart P&IDs software is not quick or time efficient.

    a. These should intelligently be able to tag encircled areas with revision / scope area information. Clouded areas and encircled elements should automatically fall under drawn cloud category e.g. revision number or scope area definitions and limitation.
    b. A drawing or diagram should be very neat and clean with eye catching standardized looks controlled under ISO, DIN or ASME standards. Therefore, aesthetically these should be nice enough to pass through required flanges and Tie-In points.
    c. There is a need to feel the importance of revision / scope clouds. Once we feel the importance then a mutually decided and well consented revision and scope clouding system may be designed.
3. A reliable and intelligently extracted line list needs excellent configuration as well as a very good team of process designers. During drafting if main / primary lines are mixed up with branches, vent

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and drain lines then it is next to impossible to extract reliable line list. Such drafting practice invites multifold re-checking and bug fixing activities. Software vendors ought to feel the importance of this element as well and provide tailor-made solutions. Few proposals are listed below:

a. Different operating companies have different templates for line, instrument, equipment and valves reporting. Smart P&ID software vendor may provide ready-made templates based on inputs from big client companies.
b. There should be provisions to restrict drafting teams within certain parameters under the umbrella of already set policies and rules. For example, if a branch is tapped out of a pipeline and is not given a separate line number then its priority should be set to secondary, automatically by smart P&ID software. Likewise, vent and drain lines should automatically be identified by the smart P&ID software to keep their identity separate and unique.
c. If the same line type is used to draw vent or drain lines which are tapping out of a main line, then the system should restrict the designer to do so.

4. After consultation with different operating companies some project templates may be configured and placed within smart P&ID setup files directories. It will help to install the software and smoothly map the operating company's specific requirements. It will produce seamless results with minimum configuration.

However, all above may be configured in all available smart P&ID software, but only an excellent smart P&ID admin can configure it. There is a need that smart P&ID software vendors to feel the importance of reporting systems for any processing plant. Finally, this all will serve as a digital twin model. Previously plant design had outputs and results in siloes and were loosely integrated for specific departments, but now digital twin requirements demand hi-level integration and centralization. Tailor-made solutions should be provided. These will make drafting easy and data integration and flow easy and straightforward.

Digital Twin and AI

Next milestone is AI deployment on any digital twin model. Once real production starts then future output predictions may be further fine-tuned after deploying AI regression, classification or cluster models. Furthermore, several studies with the help of AI may be conducted on digital twin models of any processing plant

1. Life span calculations of the processing plant.
2. Product (slurries, liquid or gas) composition elements study and analysis based on forecasted and real output.
3. Certain piping components, instrument and equipment replacement and upgradation requirements.
4. Safety studies may be done in digital atmosphere; provided live plant data is available in digital twin model.

All EPC project documents and drawings are available in digital format. With the help of modern digital techniques, these documents and drawings (2D diagrams, orthographic projection and 3D models) are available together in a single platform i.e., digital twin.

As mentioned earlier, AI may help to achieve certain goals ranging from EPC project execution to processing plant life span. AI supervised, un-supervised or reinforcement models are data hungry and need data. Data may easily be taken from digital documents or drawings (2D and 3D models).

Forestalling Data Loss / Inaccessibility Issues in Future

Real life processing plants life span is around 03 to 05 decades. Experience witnesses that digital assets are prone to inaccessibility after 02 to 03 decades.

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Severally, it is observed that it is impossible many times to access a 10 year old digital file. Same is applicable to digital twin.

1. If digital twin software vendor is out of business within next 02 decades, then what is the fate of digital twin model?
2. During the project life span much upgradation, modifications, demolition and revamping is done. How to reflect the changes in the digital twins, from time to time?
3. If a new software vendor jumps in with some excellent solutions, then how to retrieve / migrate legacy digital twin in the new setup?
4. Processing plant documents, drawings and all digital assets including digital twin model will be in a very bad situation if supporting software is out of the market due to any unforeseen reason.
5. AI is data hungry but proprietary rights driven digital twin models and applications will not provide an easy access to its whole system. Proprietary products codes and internal structure are mostly encrypted or visible in machine language only. Without specific software, adopters, patches or plug-ins accessing data from them is next to impossible. Data utilization from EPC documents is relatively easy because we can channelize or neutralize documents data with TXT, CSV or XML file formats, but neutralizing data from 2D and 3D models is very cumbersome and all available CAD software has not any common file formatting system to address this issue.

As of now, any digital file, systems or database is bound to be accessed via some proprietary software. During digital twin creation several software vendor applications are used and then data from them is tunneled via gateways, adaptors or plug-ins to neutralize the data under certain ISO standards and then in the end we again pass same to a proprietary software i.e. digital twin.

Proposed Solutions

CFIHOS (Capital Facilities Information Handover Specifications), ISO 8601 and ASCII-based format (American Standard Code for Information) ISO 10303 standards may be strengthened, involved and implemented more forcefully to make 2D and 3D data exchange more neutral and accessible via simple text editors even.

1. To deal with 2D and 3D CAD data DXF and STEP are open-source file formats available to everybody under GNU or Apache licenses (General Public License).
2. Similar to DXF or STEP or better file formats may be developed or explored to make 2D and 3D data exchange possible in more neutral formats, which should be human as well as machine-readable formats.
3. Any proprietary digital twin software or related applications should be legally and professionally bound by global industrial and software communities and conglomerates to enable their software to produce above discussed ASCII-based neutral files at any required time.
4. Above stated ASCII-based neutral files may also be updated / generated parallel to normal save-as or export / import options.

Above will forestall most discussed issues in a more amicable way. No one is a loser in this proposed plan. It is a win-win situation for everybody e.g., operating companies, EPC contractors, end users and software vendors.

1. At the end client company is the owner of data. Therefore, his ownership should be true ownership without any strings attached.
2. Client should be free to jump to any other better digital twin platform or applications, whenever they want or find better options in the future.

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3. Any old or new vendor may jump in and help esteemed clients migrate their existing digital twin model and related applications data seamlessly into a new, more robust, cost-effective, efficient, secure, fault-proof and better platform easily.
4. It will open the door for healthy competition to develop better digital twin applications and technology in the future.
5. Most of the time, AI needs ASCII-based data to train its image processing, NLP (Natural Language Processing) or tabular data related supervised, un-supervised or reinforcement models.
6. So far, data in an ACSII-based file format is a promising data migration / utilization format for any AI models training. More plugins and adopter's involvement to get neutralized files will obviously generate messy and unclean data. Garbage-in and garbage-out is the primary principle to get a good AI trained model for accurate forecasting and analysis.

The above stated proposals will pave a way for excellent data acquisition to train good AI models.

Conclusion

In any digital twin model P&ID importance is evident therefore upgraded smart P&ID software, which is considering P&ID importance better than before from all aspects and tackle brown field P&IDs issues e.g., demolition and construction P&IDs nicely in any processing plant database have multifold benefits. In the later stages of any running processing plant when revamping is required, then revision and scope clouds intelligent entry in the smart P&IDs system and tailor-made solutions for reporting system carry more benefits for digital twin goals as well as for any AI model training.

EPC projects execution range life from 03 to 05 years, but on other end processing plant real life span range from 30 to 50 years. There is a need to work on concrete footings to evolve digital twin models and applications.

1. Digital twin life span should not be haunted by the ghost of mortality e.g., inaccessibility after 10 or more years or unavailable data links to get required data easily from related applications for any kind of AI models training.
2. Proposed parallel ASCII-based data saving / transfer mechanism is the need of hour to make digital twin model and applications immortal and available at any time and for anybody easily.

AI is the talk of the day and next milestone in the foreseeable future. Processing plants may autonomously operate, deliver live and real data with the help of AI driven robots, systems and available cloud based massive computing and processing power. There is a need to take concrete steps towards technological progress, wisely and futuristically.

References

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